

SOFTWARE ENGINEER

Brandon Temple

Software engineer focused on workflow, data, and production web systems. Builds C#/.NET desktop apps, browser-based reporting tools, Firebase-backed web platforms, Python/SQL automation, API integrations, Excel exports, and AI-assisted internal software.

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PROFESSIONAL EXPERIENCE

Software Engineer (Data Systems) / Technology Coordinator

MSU Libraries, Mississippi State University | Starkville, MS | Aug 2024 - Present

- Build and maintain internal software for MSU Libraries and Mississippi Library Partnership operations serving 12 library systems and 76 branches/campus libraries.
- Automate reporting, file processing, metadata extraction, data cleanup, and Excel-ready staff workflows using C#, .NET, Python, Perl, SQL, JavaScript, HTML/CSS, and shell scripts.
- Developed Query, a browser-based reporting frontend with API-backed field metadata, configurable filters/forms, query history/templates, JSONL result handling, and Excel export.
- Created MetaTable, a C# WPF metadata scanner for large directory trees with real-time progress tracking and structured Excel exports.
- Translate staff and member-library requests into SirsiDynix API/SQL integrations, documented support workflows, and Excel reports while communicating technical tradeoffs clearly to nontechnical users.
- Use AI-assisted development while applying C# and C++ systems fundamentals, data structures, debugging, tests, and code review to validate generated code for correctness, performance, and maintainability.

Contract Software Engineer / Web Developer

Afternoon Adventure Learning Center | Newton, MS | 2026

- Delivered a paid client web platform for a 501(c)(3) afterschool and tutoring center, combining a public custom-domain website with family portal, attendance, QR-based staff time-clock/payroll, and staff dashboard workflows.
- Implemented Firebase Auth, Firestore, Storage, Gen 2 Cloud Functions, App Check-protected callables, Security Rules, role-based staff permissions, secure account deletion, generated PDFs/reports, traffic metrics, and email workflows.
- Worked directly with center stakeholders to translate registration, tuition, transportation, resources, events, donations, and content-management needs into maintainable software.

PROJECTS

Query

Browser-Based Report Builder | JavaScript, HTML, CSS, ExcelJS

Browser reporting frontend with API-backed field metadata, configurable filters/forms, query history/templates, JSONL result handling, virtualized tables, and Excel export.

github.com/bt1142msstate/Query

MetaTable

WPF File Metadata Scanner | C#, .NET 9, EPPlus

C# WPF scanner for large directory trees; extracts file metadata, tracks progress, and exports structured Excel reports for MSU Libraries.

github.com/bt1142msstate/MetaTable

Afternoon Adventure Website

Firebase-Backed Nonprofit Web Platform | HTML, CSS, JavaScript

Paid nonprofit web platform with Firebase Auth, Firestore/Storage, Security Rules/App Check, staff dashboard tools, attendance/guardian workflows, QR-based staff time clock, payroll CSV/PDF reports, enrollment PDF generation, family accounts, and SEO metadata.

github.com/bt1142msstate/afternoon-adventure

TECHNICAL SKILLS

Languages & UI C#, Python, Perl, SQL, JavaScript, TypeScript, HTML/CSS, XAML, C++

.NET/Desktop .NET 8/9, ASP.NET Core, WPF, WinUI 3, Windows App SDK, WebView2, MSIX Packaging

Data & APIs SQL Server, PostgreSQL, EF Core, REST APIs, JSONL/NDJSON, SirsiDynix APIs, CGI, JSON, MARC, Record Linkage

Cloud & Firebase Firebase Auth, Firestore, Firebase Storage, Cloud Functions, Firebase Security Rules, App Check, GitHub Pages

Libraries EPPlus, ExcelJS, PDF.js, PDF-LIB, Nodemailer, MetadataExtractor, ImageSharp, AutoNumeric, FFmpeg

Tools & Delivery Git, GitHub Actions, Visual Studio, VS Code, Postman, MSTest, Python unittest, ESLint, Playwright, Make, PowerShell/Shell, CI/CD

Methods & AI Data Structures, Algorithms, TDD, Agile, AI-First Engineering, Code Review, Automation, Machine Learning

EDUCATION

Master of Science in Software Engineering

Mississippi State University | Aug 21, 2024 - May 2027 (Expected)

Starkville, MS | GPA: 4.0/4.0

Bachelor of Science in Computer Science, Cum Laude

Mississippi State University | Aug 18, 2021 - May 10, 2024

Starkville, MS | GPA: 3.68/4.0

Honors: Phi Theta Kappa, Gamma Beta Phi

Associate of Arts in Computer Science

East Central Community College | Aug 12, 2019 - May 7, 2021

Decatur, MS | GPA: 3.85/4.0

Honors: Phi Theta Kappa, Gamma Beta Phi

President's List: Fall 2019, Spring 2020

Dean's List: Fall 2020

C++ Programming Award

Outstanding ECCC CS Student Award, awarded by Mississippi State University (custom cowbell)

ADDITIONAL EXPERIENCE

Community Assistant

College View, Mississippi State University | 2022 - 2024

- Supported student residents through communication, policy enforcement, community operations, and incident response.
- Served a 600+ student residential community through crisis response, resident support, and escalation coordination.
- Communicated with residents and staff during incidents, documented issues clearly, and escalated concerns through the proper support channels.

STEM Leader & Math Tutor

Afternoon Adventure Learning Center | 2018 - 2021

- Tutored students in math, science, and technology.
- Prepared students for exams and improved study strategies.
- Designed STEM activities in coordination with staff.

Computer Science Lab Monitor

East Central Community College | 2019 - 2020

- Maintained and troubleshoot lab computers and systems.
- Assisted students with technical issues and software use.
- Supported technical problem-solving and computer literacy.